# Planning and Reinforcement Learning in AI Systems

(PARLAI)

## Lesson 7b - Model-based RL (part 1)

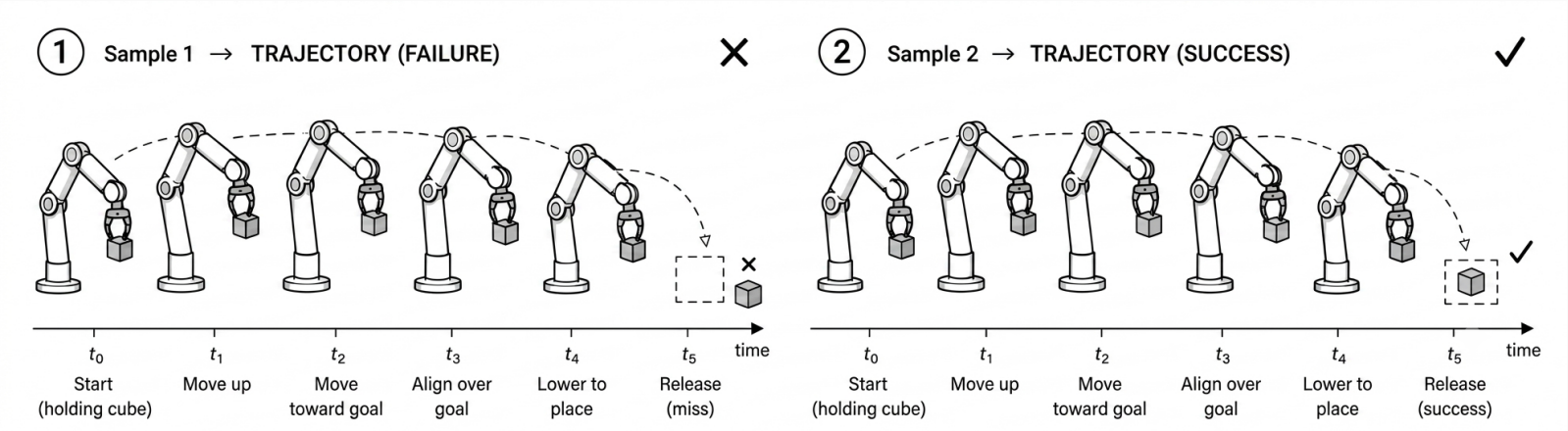
Sarah Keren

The Taub Faculty of Computer Science
Technion - Israel Institute of Technology

# Motivation

# Model-based RL Motivation

Suppose we have successfully trained a robot to grasp the cube.

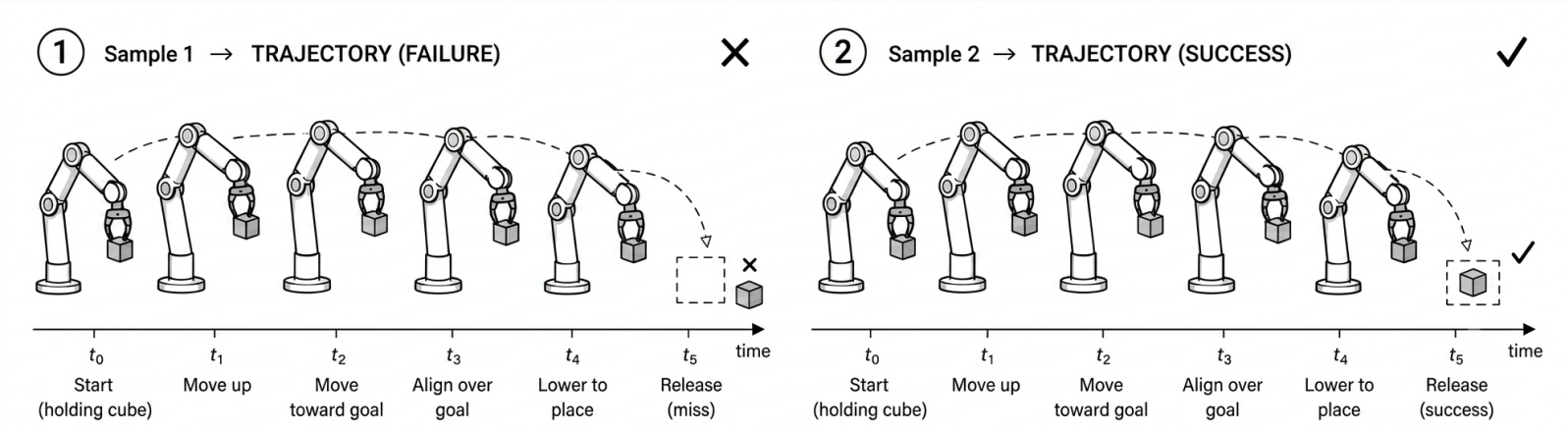




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# Model-based RL Motivation

Suppose we have successfully trained a robot to grasp the cube.




What happens if the cube is moved slightly?

What if the robot starts from a different configuration?

What if there is an obstacle in the way?

Can the agent adapt without thousands of new interactions?


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# Model-free vs. Model-based RL

Two main approaches to decision-making in RL.

- **Model-based approach:** use a predictive model of the world to ask questions of the form "what will happen if I do x?" to choose the best action.
- **Model-free approach:** bypass the modeling step in favor of **learning a control policy directly**. "how much reward will I get if I do x?"

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# Model-based RL Motivation

Model-based RL exploits a learned model to predict the consequences of actions, enabling planning and generalization to new situations with fewer real-world interactions.




Although in practice the line between these two techniques can become blurred, as a coarse guide it is useful for dividing up the space of algorithmic possibilities.

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# Advantages and Disadvantages of Model-Based RL

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# Advantages and Disadvantages of Model-Based RL

Advantages:

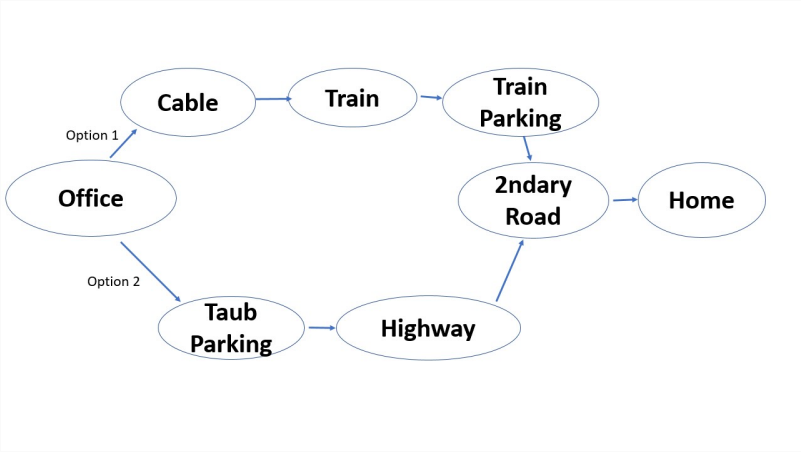
- Can be more sample efficient
- Model learning can use supervised learning methods
- Can support planning and counterfactual reasoning
- Can reason about model uncertainty
- Often more explainable

Disadvantages:

- Model errors can lead to bad plans
- Planning with the model can be computationally expensive
- Long simulated rollouts may suffer from compounding errors
- Learning an accurate model can be difficult in complex domains

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# Model-Based RL for the Going Home Example?




Ideas for model-based evaluation ?

Ideas for model-based control?

# Learning a Model

# What is a Model?

A model is a mathematical description of how the environment behaves. In fully observable RL, the model is typically represented as an MDP

$$\mathcal{M}_\eta = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}_\eta, \mathcal{R}_\eta \rangle,$$

where $\eta$ denotes the model parameters.

States $\mathcal{S}$ and actions $\mathcal{A}$ are usually assumed to be known. The goal is therefore to learn:

- How does the state change after taking an action?
- What reward does an action produce?

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# What Does the Model Predict?

Mathematically, we approximate the transition and reward functions:

$$S_{t+1} \sim \mathcal{P}_\eta(S_{t+1} \mid S_t, A_t)$$

$$R_{t+1} \sim \mathcal{R}_\eta(R_{t+1} \mid S_t, A_t)$$

A common simplifying assumption is that the next state and reward are conditionally independent given the current state and action:

$$\mathbb{P}(S_{t+1}, R_{t+1} \mid S_t, A_t) = \mathbb{P}(S_{t+1} \mid S_t, A_t)\,\mathbb{P}(R_{t+1} \mid S_t, A_t).$$

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# Model Learning

Our goal is to estimate model $\mathcal{M}_\eta$ from experience collected by interacting with the environment $\{S_1, A_1, R_2, \ldots, S_T\}$

This is a supervised learning problem where the data set consists of interactions

$$S_1, A_1 \rightarrow R_2, S_2$$

$$S_2, A_2 \rightarrow R_3, S_3$$

$$\vdots$$

$$S_{T-1}, A_{T-1} \rightarrow R_T, S_T$$

What are design decisions we need to make?

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# Reminder: Supervised Learning Components

A **distribution** $\mathcal{D}$, from which samples are drawn i.i.d. (independent and identically distributed)

A **sample**, or labeled example $(x_i, y_i)$ is a pair consisting of:

- an **input** (feature vector) $x_i$
- the corresponding **label** or target value $y_i$

A **sample set**: $\{(x_i, y_i)\}_{i=1}^m$ is used to learn a **predictor** $h(x) \in H$ from a **hypothesis class** $H$ that generalizes well with respect to the underlying distribution $\mathcal{D}$, i.e., predicts the correct label for new inputs $(x, y) \sim \mathcal{D}$ with high probability.

When the label $y$ takes values in a **discrete set** (e.g., $\{0, 1\}$ or a set of classes), the task is called **classification**.

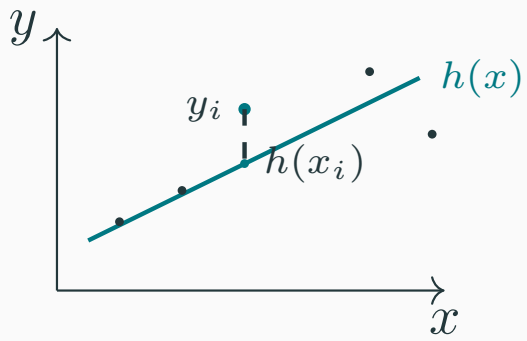
When the label $y$ takes values in a **continuous space** (e.g., $y \in \mathbb{R}$), the task is called **regression**.

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# Reminder: Loss and Risk

In supervised learning, we aim to **learn a hypothesis that minimizes prediction error on unseen data**.

**Loss** $\ell(h(x), y)$: point-wise error on a **single example**




**Expected Risk** $\mathcal{R}(h)$: expected loss over the **data distribution**

$$\mathcal{R}(h) = \mathbb{E}_{(x,y)\sim\mathcal{D}}[\ell(h(x), y)]$$

Since we typically do not have access to the true distribution $\mathcal{D}$, we rely instead on a **finite dataset** $(x, y) \sim \mathcal{D}$ to compute the **Empirical Risk** $\hat{\mathcal{R}}(h)$, i.e., the average loss over the **training data**.


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# Reminder: Empirical Risk Minimization

Given a sample set $\{(x_i, y_i)\}_{i=1}^m$ and a hypothesis class $H$, we evaluate each hypothesis by its average loss on the training data.

For a hypothesis $h \in H$, the **empirical risk** is

$$\hat{\mathcal{R}}(h) = \frac{1}{m} \sum_{i=1}^{m} \ell(h(x_i), y_i)$$

The learning problem is to find a hypothesis with minimal empirical risk:

$$\hat{h} = \arg\min_{h \in H} \hat{\mathcal{R}}(h) = \arg\min_{h \in H} \frac{1}{m} \sum_{i=1}^{m} \ell(h \in h(x_i), y_i)$$

**Interpretation:** choose the predictor whose predictions incur the smallest average loss on the observed examples.

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# Reminder: Examples of Loss Functions

## Classification

- 0-1 Loss

$$\ell(h(x), y) = \begin{cases} 0 & h(x) = y \\ 1 & h(x) \neq y \end{cases}$$

- Cross-Entropy Loss (used with probabilistic classifiers)

$$\ell(h(x), y) = -\log P(y|x)$$

## Regression

- Squared Loss (Mean Squared Error)

$$\ell(h(x), y) = (h(x) - y)^2$$

- Absolute Loss

$$\ell(h(x), y) = |h(x) - y|$$

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# Reminder: Supervised Learning Setup

**Input:** A finite data set generated from an unknown distribution $(x, y) \sim \mathcal{D}$ over inputs and labels, and a hypothesis class $H$

The data set is assumed to be sampled **independently and identically distributed** from some underlying distribution $\mathcal{D}$.

**Optimality:** a hypothesis that minimizes the expected loss:

$$h^\star = \arg\min_{h \in H} \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\ell(h(x), y)\right]$$

**Empirical Objective:** Since the true distribution $\mathcal{D}$ is unknown, we approximate this objective using the empirical loss:

$$\hat{h} = \arg\min_{h \in H} \frac{1}{m} \sum_{i=1}^{m} \ell(h(x_i), y_i)$$

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# Reminder: Supervised Learning Setup

In practice, we split the available dataset into three parts:

**Training set** $D_{\text{train}}$:

- used to **learn the model parameters**
- optimize empirical risk:

**Validation (evaluation) set** $D_{\text{val}}$:

- used to **tune hyperparameters** (e.g., model size)
- helps detect **overfitting**

**Test set** $D_{\text{test}}$:

- used **only once at the end**
- provides an **unbiased estimate of generalization performance**

Never use test data during training or model selection.

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# Back to model learning

# Model Learning: Design Decisions

**Model Representation:** What model class should we use?

table, linear model, Gaussian process, neural network, etc.

**Model Output:** Should the model predict a value or a distribution?

**Measuring Loss:** What loss function should we use?

MSE, negative log-likelihood, KL divergence, etc.

There are many ways to answer these questions; we will cover only some possible solutions

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# Model Representation

## Examples of Models

- Lookup table model
- Linear expectation model
- Linear Gaussian model
- Gaussian process model
- Deep belief network model
- ...




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# Model Representation: Lookup Table

A lookup table model stores an explicit estimate of the MDP:

$$\hat{\mathcal{M}} = \langle \mathcal{S}, \mathcal{A}, \hat{\mathcal{P}}, \hat{\mathcal{R}} \rangle.$$

For each state-action pair $(s, a)$, the table stores the estimated transition probabilities and expected reward:

$$(s, a) \quad \longmapsto \quad \hat{\mathcal{P}}(\cdot \mid s, a), \quad \hat{\mathcal{R}}(s, a).$$

A 3-state 2-action example:

| $s$ | $a$ | $\hat{\mathcal{P}}(\cdot \mid s, a)$ | $\hat{\mathcal{R}}(s, a)$ |
|---|---|---|---|
| $s_1$ | $a_1$ | $[0.1,\ 0.7,\ 0.2]$ | $3.4$ |
| $s_1$ | $a_2$ | $[0.6,\ 0.2,\ 0.2]$ | $1.1$ |
| $s_2$ | $a_1$ | $[0.0,\ 0.8,\ 0.2]$ | $5.0$ |
| | | $\dots$ | |

How do we fill the table?

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# Model Representation: Lookup Table

Idea: Count experiences.

$$N(s,a) = \sum_{t=1}^{T-1} \mathbf{1}\{S_t = s, A_t = a\}$$

$$\hat{\mathcal{P}}(s' \mid s,a) = \frac{\sum_{t=1}^{T-1} \mathbf{1}\{S_t = s, A_t = a, S_{t+1} = s'\}}{N(s,a)}$$

$$\hat{\mathcal{R}}(s,a) = \frac{\sum_{t=1}^{T-1} \mathbf{1}\{S_t = s, A_t = a\} R_{t+1}}{N(s,a)}$$

Alternatively:

- At each time-step $t$, record experience tuple $\langle S_t.A_t, R_{t+1}, S_{t+1} \rangle$
- To sample model, randomly pick tuple matching $\langle s, a, \cdot, \cdot \rangle$

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# Model Output: Deterministic vs Probabilistic

A deterministic model predicts a single next state:

$$\hat{s}_{t+1} = f_\eta(s_t, a_t).$$

A probabilistic model predicts uncertainty:

$$p_\eta(s_{t+1} \mid s_t, a_t).$$

This is useful because the agent can distinguish predictions it is confident and uncertain about.

Knowing that the model is uncertain can be as important as the prediction itself.

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# Model Output: Regression vs Density Estimation

Different parts of the model may be learned in different ways.

Regression predicts an expected value. Density estimation predicts how likely each possible outcome is.

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# Model Output: Regression vs Density Estimation

Different parts of the model may be learned in different ways.

Regression predicts an expected value. Density estimation predicts how likely each possible outcome is.

**Reward prediction** is often treated as regression:

$$\mathcal{R}_\eta(s, a) \approx \mathbb{E}[R_{t+1} \mid s, a].$$

The model predicts the expected reward.

**Transition prediction** is often treated as density estimation:

$$\mathcal{P}_\eta(s' \mid s, a) \approx \mathbb{P}(S_{t+1} = s' \mid s, a).$$

The model predicts a probability distribution.

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# Measuring Loss

To learn the model parameters $\eta$, we choose a loss function and minimize empirical error over the collected data.

$$\eta^* = \arg\min_{\eta} \sum_{t=1}^{T-1} \ell_t(\eta)$$

The loss function depends on what the model predicts: values or distributions

Common choices:

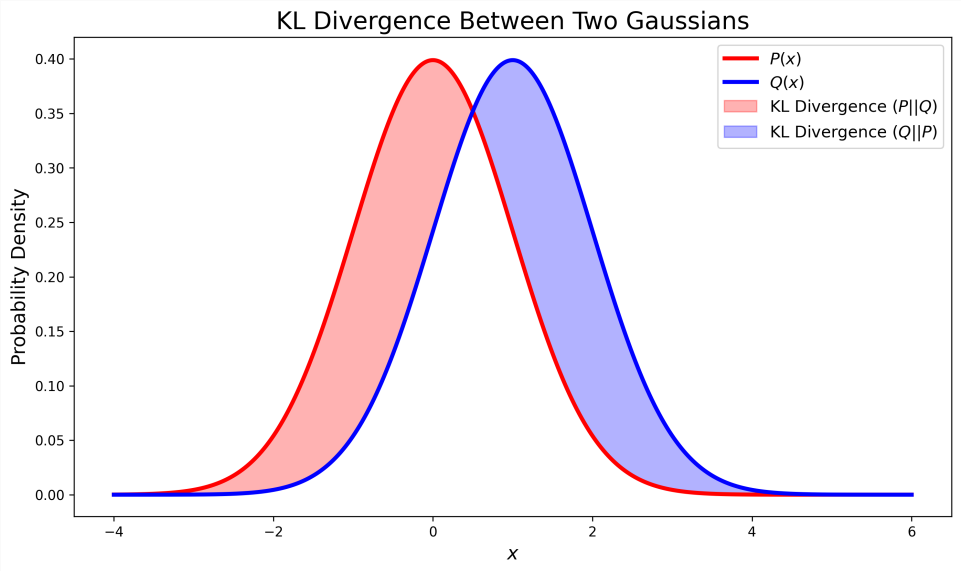
- **Mean-squared error** for deterministic prediction:
  $$\ell_t(\eta) = \|S_{t+1} - f_\eta(S_t, A_t)\|^2 .$$
- **Negative log-likelihood** for probabilistic prediction:
  $$\ell_t(\eta) = -\log p_\eta(S_{t+1} \mid S_t, A_t).$$
- **KL divergence** for comparing full probability distributions:
  $$\ell_t(\eta) = D_{\mathrm{KL}}\left(p(\cdot \mid S_t, A_t) \,\|\, p_\eta(\cdot \mid S_t, A_t)\right).$$

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# KL Divergence

**Divergence** measures how different two probability distributions are. **KL divergence** measures how much information is lost when we use an approximate distribution $q$ instead of the true distribution $p$:

$$D_{\mathrm{KL}}(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$





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# KL Divergence

Suppose the true next-state distribution is

$$p(s_{t+1} \mid s_t, a_t),$$

and our learned model predicts

$$q_\eta(s_{t+1} \mid s_t, a_t).$$

To measure how different these two distributions are, we use the **Kullback–Leibler (KL) divergence**:

$$D_{\mathrm{KL}}(p\|q) = \sum_{s'} p(s') \log \frac{p(s')}{q(s')}.$$

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# KL Divergence

Properties:

- $D_{\mathrm{KL}}(p\|q) \geq 0$
- $D_{\mathrm{KL}}(p\|q) = 0$ iff $p = q$
- It is *not* symmetric: $D_{\mathrm{KL}}(p\|q) \neq D_{\mathrm{KL}}(q\|p)$

KL divergence measures the amount of information lost when the learned model $q_\eta$ approximates the true dynamics $p$.

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# KL Divergence Intuition

KL divergence compares *probability distributions*.

Suppose the true transition probabilities are

$$p = \begin{bmatrix} 0.7 & 0.2 & 0.1 \end{bmatrix}$$

the two distributions are more similar if our learned model predicts

$$q = \begin{bmatrix} 0.6 & 0.3 & 0.1 \end{bmatrix}$$

than if the model predicts

$$q = \begin{bmatrix} 0.1 & 0.2 & 0.7 \end{bmatrix}$$

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# KL Divergence Intuition

Given, $p = \begin{bmatrix} 0.7 & 0.2 & 0.1 \end{bmatrix}$ and two learned models $q_1 = \begin{bmatrix} 0.6 & 0.3 & 0.1 \end{bmatrix}$ and $q_2 = \begin{bmatrix} 0.1 & 0.2 & 0.7 \end{bmatrix}$

$$D_{\mathrm{KL}}(p\|q_1) = 0.7 \log \frac{0.7}{0.6} + 0.2 \log \frac{0.2}{0.3} + 0.1 \log \frac{0.1}{0.1}$$
$$\approx 0.027$$

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# KL Divergence Intuition

Given, $p = \begin{bmatrix} 0.7 & 0.2 & 0.1 \end{bmatrix}$ and two learned models $q_1 = \begin{bmatrix} 0.6 & 0.3 & 0.1 \end{bmatrix}$ and $q_2 = \begin{bmatrix} 0.1 & 0.2 & 0.7 \end{bmatrix}$

$$D_{\mathrm{KL}}(p\|q_1) = 0.7 \log \frac{0.7}{0.6} + 0.2 \log \frac{0.2}{0.3} + 0.1 \log \frac{0.1}{0.1}$$
$$\approx 0.027$$

$$D_{\mathrm{KL}}(p\|q_2) = 0.7 \log \frac{0.7}{0.1} + 0.2 \log \frac{0.2}{0.2} + 0.1 \log \frac{0.1}{0.7}$$
$$\approx 1.168$$

The first learned model is closer to the true transition distribution.

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# KL Divergence

In practice, the true distribution $p$ is usually unknown.

What are possible distributions to use?

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# KL Divergence and Negative Log-Likelihood

Ideally, we would like the learned transition model to be close to the true transition distribution:

$$D_{\mathrm{KL}}\left(p(\cdot \mid s, a) \,\|\, q_\eta(\cdot \mid s, a)\right).$$

This is the distribution-level objective:

- $p(\cdot \mid s, a)$ is the true environment dynamics,
- $q_\eta(\cdot \mid s, a)$ is the learned model,
- minimizing KL means making $q_\eta$ close to $p$.

The problem is that the true distribution $p$ is unknown. We only observe samples $(S_t, A_t, S_{t+1})$ from it.

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# From KL Divergence to Log-Likelihood

For a fixed state-action pair $(s, a)$, write

$$p_{s,a}(s') = p(s' \mid s, a), \qquad q_{\eta,s,a}(s') = q_\eta(s' \mid s, a).$$

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# From KL Divergence to Log-Likelihood

For a fixed state-action pair $(s, a)$, write

$$p_{s,a}(s') = p(s' \mid s, a), \qquad q_{\eta,s,a}(s') = q_\eta(s' \mid s, a).$$

Then

$$\begin{aligned} D_{\mathrm{KL}}(p_{s,a} \| q_{\eta,s,a}) &= \sum_{s'} p_{s,a}(s') \log \frac{p_{s,a}(s')}{q_{\eta,s,a}(s')} \\ &= \underbrace{\sum_{s'} p_{s,a}(s') \log p_{s,a}(s')}_{\text{does not depend on } \eta} - \sum_{s'} p_{s,a}(s') \log q_{\eta,s,a}(s') \\ &= \text{constant} - \mathbb{E}_{s' \sim p(\cdot|s,a)} \left[ \log q_\eta(s' \mid s, a) \right]. \end{aligned}$$

Therefore, minimizing KL is equivalent to maximizing the expected log-likelihood of the true next state.


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# Interpreting Negative Log-Likelihood

Given an observed transition

$$(S_t, A_t, S_{t+1}),$$

the model is rewarded for assigning high probability to the next state that actually occurred.

$$q_\eta(S_{t+1} \mid S_t, A_t) = 0.9 \quad \Rightarrow \quad -\log(0.9) \approx 0.105$$

$$q_\eta(S_{t+1} \mid S_t, A_t) = 0.01 \quad \Rightarrow \quad -\log(0.01) \approx 4.605$$

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# Interpreting Negative Log-Likelihood

Given an observed transition

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the model is rewarded for assigning high probability to the next state that actually occurred.

$$q_\eta(S_{t+1} \mid S_t, A_t) = 0.9 \quad \Rightarrow \quad -\log(0.9) \approx 0.105$$

$$q_\eta(S_{t+1} \mid S_t, A_t) = 0.01 \quad \Rightarrow \quad -\log(0.01) \approx 4.605$$

High probability for the observed transition gives a small loss.
Low probability for the observed transition gives a large loss.

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# Theorem: NLL Minimizes KL Divergence

Let $p(s'|s,a)$ be the true transition distribution and $q_\eta(s'|s,a)$ be the learned model.

For a fixed $(s,a)$, the expected negative log-likelihood is

$$\mathbb{E}_{s'\sim p}\left[-\log q_\eta(s'|s,a)\right].$$

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For a fixed $(s,a)$, the expected negative log-likelihood is

$$\mathbb{E}_{s'\sim p}\left[-\log q_\eta(s'|s,a)\right].$$

This can be decomposed as

$$\mathbb{E}_{s'\sim p}\left[-\log q_\eta(s'|s,a)\right] = H(p) + D_{\mathrm{KL}}(p\|q_\eta).$$

Since $H(p)$ does not depend on $\eta$, minimizing NLL minimizes $D_{\mathrm{KL}}(p\|q_\eta)$.

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# Proof: NLL and KL Divergence

Start with the expected negative log-likelihood:

$$\mathbb{E}_{s' \sim p}[-\log q_\eta(s'|s,a)] = -\sum_{s'} p(s'|s,a) \log q_\eta(s'|s,a).$$

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# Proof: NLL and KL Divergence

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$$\mathbb{E}_{s' \sim p}[-\log q_\eta(s'|s,a)] = -\sum_{s'} p(s'|s,a) \log q_\eta(s'|s,a).$$

Add and subtract $\log p(s'|s,a)$:

$$-\sum_{s'} p(s'|s,a) \log q_\eta(s'|s,a) = -\sum_{s'} p(s'|s,a) \log p(s'|s,a) + \sum_{s'} p(s'|s,a) \log \frac{p(s'|s,a)}{q_\eta(s'|s,a)}.$$

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# Proof: NLL and KL Divergence Continued

The first term is the entropy of the true transition distribution:

$$H(p) = -\sum_{s'} p(s'|s,a) \log p(s'|s,a).$$

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# Proof: NLL and KL Divergence Continued

The first term is the entropy of the true transition distribution:

$$H(p) = -\sum_{s'} p(s'|s,a) \log p(s'|s,a).$$

The second term is the KL divergence:

$$D_{\mathrm{KL}}(p \| q_\eta) = \sum_{s'} p(s'|s,a) \log \frac{p(s'|s,a)}{q_\eta(s'|s,a)}.$$

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# Proof: NLL and KL Divergence Continued

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Therefore,

$$\mathbb{E}_{s'\sim p}[-\log q_\eta(s'|s,a)] = H(p) + D_{\mathrm{KL}}(p\|q_\eta).$$

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# Limitations of Negative Log-Likelihood

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# Limitations of Negative Log-Likelihood

Negative log-likelihood trains the model only on transitions that were observed in the data.

This creates several potential problems:

- rare transitions may be poorly estimated,
- the model may be inaccurate for state-action pairs not visited often,
- planning may query the model outside the training distribution,
- small model errors can compound during imagined rollouts.

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- planning may query the model outside the training distribution,
- small model errors can compound during imagined rollouts.

If the model fits the data well, does that guarantee it will be useful for control?

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# Using a Prior Model

Sometimes we have prior knowledge about the dynamics.

For example, a prior model may come from:

- a physics simulator,
- an analytical dynamics model,
- demonstrations or offline data,
- a previously trained world model,
- a foundation model.

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# Using a Prior Model

Let the prior model be $p_0(s_{t+1} \mid s_t, a_t)$, and let the learned model be $q_\eta(s_{t+1} \mid s_t, a_t)$.

We train the model using both observed data and a prior regularizer:

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# Using a Prior Model

Let the prior model be $p_0(s_{t+1} \mid s_t, a_t)$, and let the learned model be $q_\eta(s_{t+1} \mid s_t, a_t)$.

We train the model using both observed data and a prior regularizer:

$$\ell(\eta) = \sum_{t=1}^{T-1} \Big[ \underbrace{-\log q_\eta(S_{t+1} \mid S_t, A_t)}_{\text{fit observed transition}} + \underbrace{\beta D_{\mathrm{KL}}\left(q_\eta(\cdot \mid S_t, A_t) \,\|\, p_0(\cdot \mid S_t, A_t)\right)}_{\text{stay close to prior model}} \Big].$$

The likelihood term learns from data. The KL term keeps the learned model close to the prior model. The coefficient $\beta$ controls how strongly we trust the prior.

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# Model Learning: Key Take-Aways

Model learning uses experience to build an internal approximation of the environment.

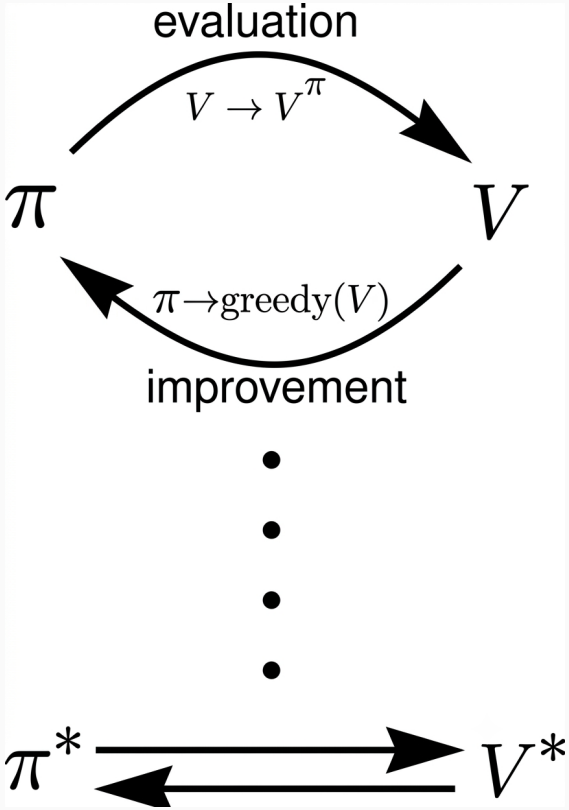
$$\text{data} \longrightarrow \text{learned model} \longrightarrow \text{decision making}$$

- With enough data, simple tabular models can estimate transition probabilities and rewards directly.
- With large or continuous state spaces, we use function approximators such as neural networks.
- The learned model is usually imperfect, so planning must account for model error and uncertainty.

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# Model-Based RL Approaches

# Reminder: Anatomy of RL Algorithms




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# Model-Based Evaluation: Adaptive Dynamic Programming (ADP)

## Adaptive Dynamic Programming (ADP)

- Follow the policy for a while
- Estimate transition model based on observations
- Learn reward function
- Use estimated model to compute utility of policy

$$V_\pi(s) = \mathcal{R}(s) + \gamma \sum_{s'} \mathcal{P}(s'|s, \pi(s) \cdot V_\pi(s')$$

How can we estimate transition model $\mathcal{P}$ and $\mathcal{R}$?

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# Model-Based Evaluation: Adaptive Dynamic Programming (ADP)

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**How can we estimate transition model $\mathcal{P}$ and $\mathcal{R}$?**

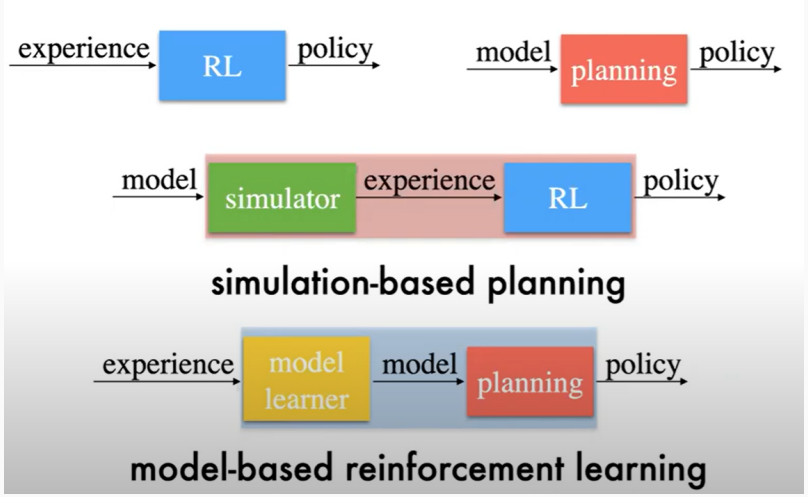
Compute the fraction of times we see $s'$ after taking $a$ in state $s$ (similarly for the reward function).

Chernoff bounds can be used to quantify the confidence in these estimates by providing high-probability bounds on the deviation of the estimated probabilities from the true probabilities.

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# Approaches to model-based control?

# Approaches to model-based control?




By Michael Littman *https://youtu.be/45FKxa3qPHo?si=chpp9SRxzjvKRZb9*

## Use the model to:

- **Create data**: generate imagined experience for learning.
- **Plan**: search, dynamic programming, or MPC.
- **Improve a policy or value function:** train the agent using model-based predictions.

# MBRL-Data Creation

# Sample-Based DM

- A simple approach: use the model only to generate samples
- Sample experiences from the estimated model

$$S_{t+1} \sim \mathcal{P}_\eta(S_{t+1}|S_t, A_t)$$

$$R_{t+1} \sim \mathcal{R}_\eta(R_{t+1}|S_t, A_t)$$

- Apply model-free RL to samples, e.g.: Monte-Carlo control, Sarsa, Q-learning

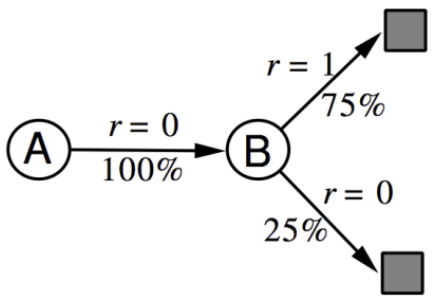
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# Sample-Based DM: Back to the AB Example

- Construct a table-lookup model from real experience
- Apply model-free RL to sampled experience

Real experience

A, 0, B, 0
B, 1
B, 1
B, 1
B, 1
B, 1
B, 1
B, 0




Sampled experience

B, 1
B, 0
B, 1
A, 0, B, 1
B, 1
A, 0, B, 1
B, 1
B, 0

e.g., Monte-Carlo learning: $V(A) = 1, V(B) = 0.75$


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# MBRL w. Planning

# DM with a Model

- Given a model $\mathcal{M}_\eta = \langle \mathcal{P}_\eta, \mathcal{R}_\eta \rangle$
- Solve the MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}_\eta, \mathcal{R}_\eta \rangle$ using favourite planning algorithm
  - Value iteration
  - Policy iteration
  - Tree search
  - Heuristic search

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# Naïve Approach To Model-Based Control

## Model-Based approach to RL

1. Act randomly for a (long) time
2. Learn transition function and reward function
3. Solve resulting MDP using value iteration, policy iteration, LAO*, MCTS etc.
4. Follow resulting policy thereafter.

Will this work ?

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Yes, if we do step 1 long enough and there are no “dead-ends”.

Any problems?

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4. Follow resulting policy thereafter.

Will this work ?

Yes, if we do step 1 long enough and there are no “dead-ends”.

Any problems?

We will act randomly for a long time before exploiting what we know

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# Revision of Naïve Approach

## Model-Based approach to RL

1. Start with initial (uninformed) model
2. Solve for optimal policy given current model (using value or policy iteration)
3. Execute action suggested by policy in current state
4. Update estimated model based on observed transition
5. Goto 2

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This is just ADP but we follow the greedy policy suggested by current value estimate

Will this work ?

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# Revision of Naïve Approach

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This is just ADP but we follow the greedy policy suggested by current value estimate

Will this work ? No. Can get stuck in local optimum

What can be done ?

# Reminder: Exploration vs. Exploitation

Two reasons to take an action in RL

**Exploitation:** We exploit our current knowledge to get a payoff.
**Exploration:** Get more information about the world. How do we know if there is not a pot of gold around the corner?

To explore, we typically need to take actions that do not seem best according to our current model.

Managing the trade-off between exploration and exploitation is a critical issue in RL.

Basic **intuition** behind most approaches:

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To explore, we typically need to take actions that do not seem best according to our current model.

Managing the trade-off between exploration and exploitation is a critical issue in RL.

Basic **intuition** behind most approaches:

- Explore more when knowledge is weak
- Exploit more as agent gains knowledge

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# ADP-based RL

## ADP-based RL

1. Start with initial model
2. Solve for optimal policy given current model (using value or policy iteration)
3. Execute action suggested by an explore/exploit policy (explores more early on and gradually uses policy from 2)
4. Update estimated model based on observed transition
5. Goto 2

This is just ADP but we follow the explore/exploit policy

**Will this work?**

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# ADP-based RL

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4. Update estimated model based on observed transition
5. Goto 2

This is just ADP but we follow the explore/exploit policy

**Will this work?** Depends on the explore/exploit policy

**Any ideas?**

# Explore/Exploit Policies

Greedy action : the action maximizing estimated Q-value

$$Q(s,a) = \mathcal{R}(s) + \gamma \sum_{s'} \mathcal{P}(s'|s,a) \cdot V_\pi(s')$$

where

- $V$ is the current optimal value function estimate (based on the current model),
- $\mathcal{R}$ and $\mathcal{P}$ are current estimates of the model, and
- $Q(s,a)$ is the expected value of taking action $a$ in state $s$ and then getting the estimated value $V(s')$ of the next state $s'$.

We want an exploration policy that is GLIE (greedy in the limit of infinite exploration).

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# Planning in a Learned Model

Assume we have learned a model

$$\hat{M} = \langle \mathcal{S}, \mathcal{A}, \hat{P}, \hat{R} \rangle.$$

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# Planning in a Learned Model

Assume we have learned a model

$$\hat{M} = \langle \mathcal{S}, \mathcal{A}, \hat{P}, \hat{R} \rangle.$$

Once the model is fixed, we can plan inside it using a Bellman operator:

$$(\hat{T}V)(s) = \max_{a \in \mathcal{A}} \left[ \hat{R}(s,a) + \gamma \sum_{s'} \hat{P}(s'|s,a) V(s') \right].$$

This is the Bellman optimality operator, but applied to the learned model.

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# Planning in a Learned Model

Theorem: Bellman Contraction for a Learned Model: Let $\hat{T}$ be the Bellman optimality operator for a fixed learned model $\hat{M}$:

$$(\hat{T}V)(s) = \max_a \left[ \hat{R}(s,a) + \gamma \sum_{s'} \hat{P}(s'|s,a)V(s') \right].$$

For any two value functions $V$ and $U$,

$$\|\hat{T}V - \hat{T}U\|_\infty \leq \gamma \|V - U\|_\infty.$$

Therefore, $\hat{T}$ is a $\gamma$-contraction.

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# Planning in a Learned Model

For a fixed state $s$,

$$|(\hat{T}V)(s) - (\hat{T}U)(s)| = \left|\max_a Q_V(s,a) - \max_a Q_U(s,a)\right|$$
$$\leq \max_a |Q_V(s,a) - Q_U(s,a)| .$$

where

$$Q_V(s,a) = \hat{R}(s,a) + \gamma \sum_{s'} \hat{P}(s'|s,a)V(s').$$

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# Planning in a Learned Model

Now compare the two $Q$-values:

$$
\begin{aligned}
|Q_V(s,a) - Q_U(s,a)| &= \gamma \left| \sum_{s'} \hat{P}(s'|s,a) \left(V(s') - U(s')\right) \right| \\
&\leq \gamma \sum_{s'} \hat{P}(s'|s,a) |V(s') - U(s')| \\
&\leq \gamma \|V - U\|_\infty .
\end{aligned}
$$

Therefore,

$$
\|\hat{T}V - \hat{T}U\|_\infty \leq \gamma \|V - U\|_\infty .
$$


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# Planning in a Learned Model

Because $\hat{T}$ is a contraction, repeated value iteration in the learned model converges:

$$V_{k+1} = \hat{T} V_k.$$

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# Planning in a Learned Model

Because $\hat{T}$ is a contraction, repeated value iteration in the learned model converges:

$$V_{k+1} = \hat{T} V_k.$$

So planning inside a fixed learned model is well behaved.

But?

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# Planning in a Learned Model

Because $\hat{T}$ is a contraction, repeated value iteration in the learned model converges:

$$V_{k+1} = \hat{T} V_k.$$

So planning inside a fixed learned model is well behaved.

But?

There is an important caveat:

The value iteration converges to the optimal value of the learned model, not necessarily the optimal value of the real environment.

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# MBRL for Enhanced DM

# MBRL for Enhanced Decision Making

So far, we saw two ways to use a learned model:

- Create data: generate imagined experience.
- Plan: solve or search inside the learned model.

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# MBRL for Enhanced Decision Making

So far, we saw two ways to use a learned model:

- **Create data**: generate imagined experience.
- **Plan**: solve or search inside the learned model.

A third option is to use the model to improve the agent's learning process.

The model does not have to directly choose the action. It can provide extra training signals for learning a better value function or policy.

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- **Create data**: generate imagined experience.
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A third option is to use the model to improve the agent's learning process.

> The model does not have to directly choose the action. It can provide extra training signals for learning a better value function or policy.

Examples:

- update a value function using simulated transitions,
- construct better value targets,
- train a policy using imagined short rollouts,
- learn useful latent representations of the environment.

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# Conclusion

Model-based RL learns and uses a model of the environment:

$$\text{experience} \longrightarrow \text{model learning} \longrightarrow \text{control}$$

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# Conclusion

Model-based RL learns and uses a model of the environment:

$$\text{experience} \longrightarrow \text{model learning} \longrightarrow \text{control}$$

- A model predicts the consequences of actions:

  $$(s_t, a_t) \longrightarrow (s_{t+1}, r_{t+1}).$$

- Learning the model can be viewed as a supervised learning problem.
- Once learned, the model can be used to simulate experience, evaluate policies, or plan new actions.

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# Conclusion

| Use of model | Main idea |
|---|---|
| Create data | Generate imagined transitions and apply model-free RL. |
| Plan | Search or solve the learned MDP to choose actions. |
| Enhance learning | Use the model to improve value targets, policy updates, or representations. |

The boundary between these approaches is often blurry: many modern methods combine all three.

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# Conclusion

Learning a model is only the first step: learned models are usually imperfect.

- Data may cover only part of the state-action space,
- Predictions may be uncertain,
- small model errors can compound over long rollouts,
- Greedy planning can prevent useful exploration.

Thus, model errors, uncertainty, and poor exploration can lead to bad decisions.

More advanced model-based RL methods must decide not only *how to learn a model*, but also how to use an imperfect model safely and efficiently.

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